

Capital structure 2 Seminar solutions (Chapter 16. Berck DeMarzo)

1. Gladstone Corporation is about to launch a new product. Depending on the success of the new product, Gladstone may have one of four values next year: \$150 million, \$135 million, \$95 million, or \$80 million. These outcomes are all equally likely, and this risk is diversifiable. Gladstone will not make any payouts to investors during the year. Suppose the risk-free interest rate is 5% and assume perfect capital markets.

a. What is the initial value of Gladstone's equity without leverage?

Now suppose Gladstone has zero-coupon debt with a \$100 million face value due next year.

b. What is the initial value of Gladstone's debt?

c. What is the yield-to-maturity of the debt? What is its expected return?

d. What is the initial value of Gladstone's equity? What is Gladstone's total value with leverage?

a. $0.25 \times \frac{150 + 135 + 95 + 80}{1.05} = \109.52 million

b. $0.25 \times \frac{100 + 100 + 95 + 80}{1.05} = \89.28 million

c. $\text{YTM} = \frac{100}{89.29} - 1 = 12\%$

expected return = 5%

d. $\text{equity} = 0.25 \times \frac{50 + 35 + 0 + 0}{1.05} = \20.24 million total value = $89.28 + 20.24 = \$109.52 \text{ million}$

2. As in Problem 1, Gladstone Corporation is about to launch a new product. Depending on the success of the new product, Gladstone may have one of four values next year: \$150 million, \$135 million, \$95 million, or \$80 million. These outcomes are all equally likely, and this risk is diversifiable. Suppose the risk-free interest rate is 5% and that, in the event of default, 25% of the value of Gladstone's assets will be lost to bankruptcy costs. (Ignore all other market imperfections, such as taxes.)

a. What is the initial value of Gladstone's equity without leverage?

Now suppose Gladstone has zero-coupon debt with a \$100 million face value due next year.

b. What is the initial value of Gladstone's debt?

c. What is the yield-to-maturity of the debt? What is its expected return?

d. What is the initial value of Gladstone's equity? What is Gladstone's total value with leverage?

Suppose Gladstone has 10 million shares outstanding and no debt at the start of the year.

e. If Gladstone does not issue debt, what is its share price?

f. If Gladstone issues debt of \$100 million due next year and uses the proceeds to repurchase shares, what will its share price be? Why does your answer differ from that in part (e)?

a. $0.25 \times \frac{150 + 135 + 95 + 80}{1.05} = \109.52 million

b. $0.25 \times \frac{100 + 100 + 95 \times 0.75 + 80 \times 0.75}{1.05} = \78.87 million

c.
$$YTM = \frac{\$100}{\$78.87} - 1 = 26.79\%$$

expected return = 5%

d.
$$\text{equity} = 0.25 \times \frac{50 + 35 + 0 + 0}{1.05} = \$20.24 \text{ million total value}$$

$$= 0.25 \times \frac{150 + 135 + 95 \times 0.75 + 80 \times 0.75}{1.05} = \$99.11 \text{ million}$$

(or $78.87 + 20.24 = \$99.11$ million)

e.
$$\frac{109.52}{10} = \$10.95 / \text{share}$$

f.
$$\frac{99.11}{10} = \$9.91 / \text{share}$$
 Bankruptcy cost lowers share price.

Note that Gladstone will raise \$78.87 million from the debt, and repurchase $\frac{78.87}{9.91} = 7.96$ million shares

. Its equity will be worth \$20.24 million, for a share price of $\frac{20.24}{10 - 7.96} = \9.91 after the transaction is completed.

3. **Kohwe Corporation plans to issue equity to raise \$40 million to finance a new investment. After making the investment, Kohwe expects to earn free cash flows of \$10 million each year. Kohwe currently has 5 million shares outstanding, and it has no other assets or opportunities. Suppose the appropriate discount rate for Kohwe's future free cash flows is 8%, and the only capital market imperfections are corporate taxes and financial distress costs.**

a. **What is the NPV of Kohwe's investment?**

b. **Given these plans, what is Kohwe's value per share today?**

c. **Suppose Kohwe borrows the \$40 million instead. The firm will pay interest only on this loan each year, and it will maintain an outstanding balance of \$40 million on the loan. Suppose that Kohwe's corporate tax rate is 40% and expected free cash flows are still \$10 million each year.**

What is Kohwe's share price today if the investment is financed with debt?

d. **Now suppose that with leverage, Kohwe's expected free cash flows will decline to \$9 million per year due to reduced sales and other financial distress costs. Assume that the appropriate discount rate for Kohwe's future free cash flows is still 8%.**

What is Kohwe's share price today given the financial distress costs of leverage?

a.
$$NPV \text{ of investment} = \frac{\$10 \text{ million}}{0.08} - \$40 \text{ million} = \$85 \text{ million}$$

b.
$$\text{Price per share} = \frac{\$85 \text{ million}}{5 \text{ million}} = \$17$$

c.
$$\text{Price per share} = \frac{\$85 \text{ million} + (0.4 \times \$40 \text{ million})}{5 \text{ million}} = \$20.20$$

d.
$$\text{Price per share} = \frac{\frac{\$9 \text{ million}}{0.08} - \$40 \text{ million} + (0.4 \times \$40 \text{ million})}{5 \text{ million}} = \$17.7$$

4. On May 14, 2008, General Motors paid a dividend of \$0.25 per share. During the same quarter GM lost a staggering \$15.5 billion or \$27.33 *per share*. Seven months later the company asked for billions of dollars of government aid and ultimately declared bankruptcy just over a year later, on June 1, 2009. At that point a share of GM was worth only a little more than a dollar.

- a. If you ignore the possibility of a government bailout, the decision to pay a dividend given how close the company was to financial distress is an example of what kind of cost?
- b. What would your answer be if GM executives anticipated that there was a possibility of a government bailout should the firm be forced to declare bankruptcy?

- a. Agency cost—cashing out
- b. By paying a dividend, executives increased the probability of bankruptcy and therefore the probability of receiving government funds. Because these government funds are funds that investors would not otherwise be entitled to, the payment of a dividend could actually *raise* firm value in this case.

5. Hawar International is a shipping firm with a current share price of \$5.50 and 10 million shares outstanding. Suppose Hawar announces plans to lower its corporate taxes by borrowing \$20 million and repurchasing shares.

- a. With perfect capital markets, what will the share price be after this announcement?

Suppose that Hawar pays a corporate tax rate of 30%, and that shareholders expect the change in debt to be permanent.

- b. If the only imperfection is corporate taxes, what will the share price be after this announcement?
- c. Suppose the only imperfections are corporate taxes and financial distress costs. If the share price rises to \$5.75 after this announcement, what is the PV of financial distress costs Hawar will incur as the result of this new debt?

- a. The same price, \$5.50/share, because financial transactions do not create value.

- b. $0.3 \times \frac{20}{10} + 5.5 = \$6.10 / \text{share}$

- c. $(6.1 - 5.75) \times 10 = \3.5 million

6. Marpor Industries has no debt and expects to generate free cash flows of \$15 million each year. Marpor believes that if it permanently increases its level of debt to \$45 million, the risk of financial distress may cause it to lose some customers and receive less favorable terms from its suppliers. As a result, Marpor's expected free cash flows with debt will be only \$14 million per year. Suppose Marpor's tax rate is 30%, the risk-free rate is 6%, the expected return of the market is 12%, and the beta of Marpor's free cash flows is 1.10 (with or without leverage).

- a. Estimate Marpor's value without leverage.

- b. Estimate Marpor's value with the new leverage.

- a. $R_{\text{Marpor}} = 6\% + 1.1(12\% - 6\%) = 12.6\%$

$$V_U = \frac{\$15 \text{ million}}{0.126} = \$119.05 \text{ million}$$

$$b. \quad V_L = \frac{\$14 \text{ million}}{0.126} + (0.3 \times \$45 \text{ million}) = \$124.61 \text{ million}$$

7. Petron Corporation's management team is meeting to decide on a new corporate strategy. There are four options, each with a different probability of success and total firm value in the event of success, as shown below:

	Strategy			
	A	B	C	D
Probability of Success	100%	80%	60%	40%
Firm Value if Successful (in \$ million)	50	60	70	80

Assume that for each strategy, firm value is zero in the event of failure.

- Which strategy has the highest expected payoff?
- Suppose Petron's management team will choose the strategy that leads to the highest expected value of Petron's equity. Which strategy will management choose if Petron currently has
 - No debt?
 - Debt with a face value of \$20 million?
 - Debt with a face value of \$40 million?
- What agency cost of debt is illustrated in your answer to part (b)?

a, b.

	A	B	C	D
Expected Payoff ($p \cdot V$)	50	48	42	32

Debt Face Value (D)	Equity Value = $p \cdot (V - D)$			
	A	B	C	D
0	50	48	42	32
20	30	32	30	24
40	10	16	18	16

Thus, the firm will choose strategy A when $D = 0$, B when $D = \$20$, and C when $D = \$40$.

- This result illustrates the agency cost of excessive risk taking with leverage. As the amount of leverage increases, the firm's equity holders prefer that the firm adopt riskier strategies, even though these riskier strategies lower the total value of the firm.
8. Consider the setting of Problem 7, and suppose Petron Corp. has debt with a face value of \$40 million outstanding. For simplicity, assume all risk is idiosyncratic, the risk-free interest rate is zero, and there are no taxes.
- What is the expected value of equity, assuming Petron will choose the strategy that maximizes the value of its equity? What is the total expected value of the firm?

- b. Suppose Petron issues equity and buys back its debt, reducing the debt's face value to \$5 million. If it does so, what strategy will it choose after the transaction? Will the total value of the firm increase?
- c. Suppose you are a debt holder, deciding whether to sell your debt back to the firm. If you expect the firm to reduce its debt to \$5 million, what price would you demand to sell your debt?
- d. Based on your answer to (c), how much will Petron need to raise from equity holders in order to buy back the debt?
- e. How much will equity holders gain or lose by recapitalizing to reduce leverage? How much will debt holders gain or lose? Would you expect Petron's management to choose to reduce its leverage?

a. With \$40 million face value of debt, the firm will choose strategy C. Thus, the expected value of the firm is \$42 million and the expected value of equity is \$18 million.

- b. With \$5 million in debt outstanding, management will choose strategy A:

$$\text{Equity Value} = p^*(V - D)$$

Debt Face Value (D)	A	B	C	D
5	45	44	39	30

- c. After the transaction, the debt will be risk free. Price will thus be face value.
- d. Petron will need to raise \$35 million from equity holders (because the debt is risk free and has a face value of \$35 million).
- e. The expected value of equity after the transaction is \$45 million, but equity holders had to contribute \$35 million to buy back the debt. Thus, the net expected payoff to equity holders is \$10 million. If they had not done the buyback, equity would have been worth \$18 million. Thus, equity holders lost \$8 million as a result of the buyback.

Debt holders received \$35 million and still hold \$5 million in risk-free debt, for a total payoff of \$40 million after the buyback. Had Petron not done the buyback, the debt would have been worth $60\% \times \$40 \text{ million} = \24 million . Thus, debt holders gained \$16 million as a result of the buyback.

As we can see, even though the total value of the firm increases by \$8 million by reducing leverage and eliminating agency costs, the debt holders capture more than \$8 million, and thus equity holders lose. Therefore, even though Petron has an inefficiently high level of leverage, the ratchet effect of debt overhang implies that it will not choose to reduce its leverage.